Project 1 – Edition A

Company: Apple Inc. (NASDAQ: AAPL)
Valuation date: September 1, 2026
Decision requested: Whether an investment committee should authorize further diligence toward an initial public-equity position in Apple.
Initial view: **Watch / defer.** Apple has demonstrated durable scale, strong profitability and growth in Services, but an initiate decision requires current valuation inputs and a more explicit assessment of product-cycle, China, regulatory and capital-allocation risks.

> **Prior-work disclosure:** This preliminary company baseline builds on my existing Apple course files and is not a final valuation or recommendation. Those working files were consulted for continuity of historical facts, forecast structure, sensitivity planning, and previously documented validation work. Current filing facts in this document are independently identified below and take precedence where they differ.
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> **Existing files consulted:** [Apple FY2025 filing facts](apple-2025-10k-filing-facts.md) · [Apple pro-forma model](apple_proforma.py) · [Apple sensitivity plan](apple_sensitivity_plan.md) · [Locked changed-input record](apple_lab10_locked_record.md) · [Sensitivity output](apple_lab10_sensitivity_output.md)

Primary filing and financial evidence

Primary sources: Apple Inc.'s [Form 10-Q for the quarter ended June 27, 2026](https://www.sec.gov/Archives/edgar/data/320193/000032019326000020/aapl-20260627.htm), filed July 31, 2026 (accession **0000320193-26-000020**), and its [Form 10-K for fiscal year ended September 27, 2025](https://www.sec.gov/Archives/edgar/data/320193/000032019325000079/aapl-20250927.htm), filed October 31, 2025 (accession **0000320193-25-000079**). The 10-Q is the latest filed financial information available as of the September 1, 2026 valuation date. All financial figures below are in **U.S. dollars and millions**, unless stated otherwise.

Required evidence	FY2025 observation	Filing location	Investment relevance
Entity	Apple Inc.	Cover page	Legal registrant and issuer of AAPL common stock.
Reporting period	Fiscal year ended September 27, 2025	Cover page	Establishes the financial-information date.
Filing date	October 31, 2025	EDGAR filing index	Establishes public availability of the source.
Currency / units	U.S. dollars; financial statements in millions; shares in thousands (except per-share data)	Consolidated Statements of Operations, p. 29	Prevents unit errors in valuation work.
Cash and cash equivalents	$39,544	Q3 2026 Form 10-Q, Condensed Consolidated Balance Sheets, p. 2	Cash is an enterprise-to-equity bridge asset, subject to the treatment of restricted cash and operating-cash needs.
Commercial paper	$1,997	Q3 2026 Form 10-Q, Condensed Consolidated Balance Sheets, p. 2	Short-term interest-bearing borrowing; included in debt under the stated bridge convention.
Current term debt	$11,007	Q3 2026 Form 10-Q, Condensed Consolidated Balance Sheets, p. 2	Current portion of interest-bearing debt.
Non-current term debt	$71,340	Q3 2026 Form 10-Q, Condensed Consolidated Balance Sheets, p. 2	Long-term interest-bearing debt.
Total reported term debt	$82,347	Q3 2026 Form 10-Q, Note 6, “Term Debt,” p. 10	Carrying amount before adding commercial paper.
Total debt used in initial bridge	$84,344	$1,997 commercial paper + $82,347 term debt	Explicitly includes short-term commercial paper and term debt; excludes operating leases pending class convention.
Share-count evidence	14,594,180,000 common shares outstanding as of July 17,

 2026; nine-month weighted-average diluted shares of 14.750 billion | Q3 2026 Form 10-Q, cover page; Note 3, p. 6 | The point-in-time count supports market capitalization; diluted weighted-average shares remain relevant for EPS. The valuation must state which convention it uses. |
| Operating-value driver | Nine-month Services net sales: $91,728, up 14% year over year | Q3 2026 Form 10-Q, MD&A, “Products and Services Performance,” p. 15 | Services growth and mix are potentially important to recurring revenue and margin durability. |
| Contextual operating scale | Nine-month total net sales: $364,357, up 16% year over year; Q3 total gross margin: 50.1% | Q3 2026 Form 10-Q, MD&A, pp. 14–16 | Frames current growth and operating leverage, but a full-year forecast must not annualize one quarter without justification. |

Enterprise-to-equity bridge check

This market-implied Apple bridge uses the September 1, 2026 closing price and the latest share-count and balance-sheet information publicly available by that date. It is a reconciliation check—not a DCF output—and excludes marketable securities and operating leases pending a documented policy decision.

Bridge item	$ millions	Treatment
Market-implied enterprise value	4,789,806	`Equity value + debt − cash`; USD millions.
Less: commercial paper	(1,997)	Q3 2026 Form 10-Q balance sheet; June 27, 2026.
Less: term debt	(82,347)	Q3 2026 Form 10-Q, Note 6; June 27, 2026.
Add: cash and cash equivalents	39,544	Q3 2026 Form 10-Q balance sheet; June 27, 2026.
Equals: implied equity value	4,745,006	`4,789,806 − 1,997 − 82,347 + 39,544`; USD millions.
Shares outstanding	14,594.180	14.594 billion shares as of July 17, 2026, from the Q3 2026 Form 10-Q cover page.
Implied value per share	$325.13	`4,745,006 ÷ 14,594.180`; matches the September 1, 2026 closing price.
Price source / reconciliation variance	$325.13 / $0.00	Yahoo Finance historical close on September 1, 2026; rounding to whole $ millions creates immaterial differences.

Bridge convention: Debt is commercial paper plus term debt; cash means reported cash and cash equivalents. Marketable securities, operating leases, pension obligations, preferred equity, and noncontrolling interests are excluded from this preliminary reconciliation and must be explicitly evaluated in the final DCF bridge.

Initial thesis

Apple merits continued research, not an immediate recommendation. Through the first nine months of fiscal 2026, net sales grew 16% year over year and Services net sales grew 14%; the Q3 2026 filing also reported a 50.1% total gross margin for the quarter. This supports a hypothesis of substantial ecosystem-driven earnings durability. However, the filing identifies business exposures that could materially affect normalized growth and margins, including dependence on new-product launches, intense competition, supply-chain concentration, foreign-exchange and tariff effects, and regulatory/legal developments. The current market price, required return and forward operating assumptions have not yet been evaluated; therefore, value relative to price is unknown.

Consequential assumptions

- Services revenue can continue to grow faster than total company revenue without a material deterioration in economics or regulatory constraints.

- The current product installed base and ecosystem sustain customer retention and premium pricing through future device cycles.
- The selected debt convention—commercial paper plus term debt—matches the course bridge convention. Operating leases and marketable securities are excluded until that convention is confirmed.
- The valuation will use fully diluted shares appropriate to the valuation date; the FY2025 weighted-average diluted share count is evidence, not automatically the final point-in-time share count.
- A valuation completed as of September 1, 2026 will use only information public by that date, including the July 31, 2026 Form 10-Q, rather than treating FY2025 filing data as current by itself.

Unknowns and decision-critical questions

- What is Apple’s market capitalization and enterprise value on the valuation date?
- What forward growth, operating margin, reinvestment and discount-rate assumptions are required to justify the market price?
- How durable is Services growth under App Store, advertising and platform regulation?
- What are the downside effects of China demand weakness, tariffs and supply-chain disruption?
- Under the course convention, should marketable securities, lease liabilities, restricted cash, preferred interests or noncontrolling interests be included in the enterprise-to-equity bridge?

Research plan

1. Establish valuation-date market capitalization, debt, cash and diluted-share inputs; reconcile all bridge components to the course convention.
2. Build a three-case operating forecast separating Products and Services, with explicit revenue growth, gross-margin and operating-expense assumptions.
3. Analyze Services economics and regulatory exposure, including the sensitivity of revenue and margin to App Store and advertising changes.
4. Evaluate geographic and supply-chain downside cases, with particular attention to Greater China demand, tariff costs and component concentration.
5. Value Apple using a DCF and a market-based cross-check; compare implied value per diluted share with the valuation-date price and identify the assumptions that drive the recommendation.

Project 1 – Edition B: Research Evolution and Current Position

Company: Apple Inc. (NASDAQ: AAPL)
Original Edition A valuation date: September 1, 2026
Current research cut-off: September 17, 2026
Intended user: Buy-side investment committee with no current Apple position

Current action: **Watch / defer; do not initiate at $337.00 per share.**

> **Status:** This is Edition B, prepared after Edition A. It does not alter or replace `project-1-edition-a.md` or its submitted PDF. The Final Build Update below supersedes earlier interim valuation numbers in this document.

Final Build Update – October 16, 2026

The earlier $224.42 shortcut DCF has been replaced by `apple_fcff_model.py`, an articulating FY2026E–FY2030E three-statement FCFF model. The base output is **$131.61/share** (EV $1,858,504M; equity value $1,920,677M) using 6% revenue growth, 47.0% gross margin, 15.0% operating expenses/revenue, 3.0% capex/revenue, 9.0% WACC, and 2.5% terminal growth. The bridge adds Q3 FY2026 cash plus marketable securities of $146,517M and subtracts $84,344M of debt; it uses 14,594.18M shares. Each forecast year balances and its cash-flow statement ties to cash. The causal low/base/high values are $87.07/$131.61/$185.85.

A pre-set peer policy is documented in `peer-policy.md`. The P/E cross-check remains $206.86–$239.69, while Alphabet's documented EV/revenue cross-check is 10.7747x; neither is averaged into the DCF conclusion. `app.py` is the semi-adjustable product: it has bounded controls, base reset, cases, and visible failed-input/statement checks. The README-only cold run passed on its first attempt; see `docs/cold-run.md`.

The recommendation remains watch/defer because all reported valuation checks remain below the $337.00 market price. The only uncompleted evidence item is the student-authored AI-off prediction/timestamp/commit in the locked changed-input record; this cannot be truthfully supplied by a tool. See `docs/validation-and-ai-use.md`.

Current decision

I recommend that the committee **watch/defer** Apple rather than initiate a position at the September 17, 2026 closing price of **$337.00**. The current FCFF DCF estimates **$224.42 per diluted share** under a 7% five-year FCFF-growth path, 8% WACC, and 3% terminal growth. A limited, explicitly provisional P/E cross-check using Microsoft and Alphabet implies **$206.86–$239.69** per Apple share, with a $223.28 median. I do not average these methods: they answer different questions and the peer set has material business-model limitations.

This is not a claim that the market is wrong. Under the current reverse-DCF framing, the September 17 price requires substantially faster explicit cash-flow growth than the 7% base case. The recommendation changes only if the committee obtains evidence that supports a more durable cash-flow and margin path, accepts a different but documented valuation convention, or can buy at a price that supplies a margin of safety against the valuation range.

A-to-B evidence table

Item	Edition A	Edition B	Primary driver of change	Evidence or test justifying the change or stability
Thesis / recommendation	Watch/defer because valuation relative to price was unknown.	Watch/defer because the current base DCF and provisional peer check are below the observed price.	Own analysis; course instruction; AI-assisted res

earch support. | `AAPL_reverseDCF_9-10-26.md`; `AAPL_peer_comparison_2026-09-10.md`. |
Data / source choice	FY2025 10-K and Q3 FY2026 10-Q identified as primary evidence.	Added valuation-date prices, FCFF/bridge inputs, peer annual GAAP EPS, and a dated source log.	Own analysis.	`sources.md`; Q3 FY2026 10-Q; price locators cited in the DCF and peer files.
Assumption / model design	Planned a five-year forecast and DCF; assumptions were not yet specified.	FCFF DCF uses $131.0B starting FCFF, 7% explicit FCFF growth, 8% WACC, 3% terminal growth, $146.517B cash plus marketable securities, $84.344B debt, and 14.594B shares.	Own analysis; course instruction.	`AAPL_reverseDCF_9-10-26.md`; executable `dcf.py`.
Model architecture correction	The existing Apple workbench was described as a valuation model.	The workbench is classified as **FCFE** and cannot be the final Project 1 valuation engine, which must be FCFF. Its sensitivity evidence is retained, but final valuation work must reconcile the pro-forma to FCFF.	Course instruction; own model review.	`apple_proforma.py` labels and calculates FCFE; Project 1 brief requires FCFF.
Risk / limitation	Product cycle, China, regulation, supply chain, FX, and valuation inputs remained open questions.	Peer comparability, FCFF starting-point construction, terminal-value dependence, and bridge conventions are identified as specific limitations.	Own analysis; AI-generated skeptical review, independently assessed.	`AAPL_peer_comparison_2026-09-10.md`; `AAPL_reverseDCF_9-10-26.md`; `apple_lab10_sensitivity_output.md`.

Current valuation evidence

FCFF DCF and reverse-DCF result

The FCFF DCF uses a five-year 7% growth path, 8% WACC, and 3% terminal growth. It produces **$224.42 per diluted share**; the terminal value represents **80.17%** of enterprise value. The WACC–terminal-growth grid spans $165.96–$360.32 per share, but this grid alone is not the required causal low/base/high case. The September 17 market price of $337.00 is above this base value. At that price, the reverse DCF indicates that the model needs approximately a **+9.955 percentage-point** uniform increase to each of the five explicit FCFF-growth assumptions, holding the starting FCFF, WACC, terminal growth, non-operating assets, debt, and shares fixed.

The result should be read as a required assumption under this model—not as proof that Apple is overvalued. It is most useful as a diligence question: what evidence supports a cash-flow trajectory materially above the base case?

Comparable-company cross-check

Microsoft and Alphabet are **qualified leads**, not final peers. Their P/E multiples imply an Apple range of **$206.86–$239.69**, median **$223.28**, using September 17 closing prices and annual GAAP diluted EPS. Microsoft has much higher enterprise software and cloud exposure; Alphabet relies much more heavily on advertising. Their mismatch to Apple’s hardware-led ecosystem is a constraint on the transferability of their multiples, not a reason to adjust the result toward a desired price.

Sensitivity evidence

The existing workbench’s one-at-a-time sensitivity tests show that, over its chosen ranges, revenue growth is more consequential than gross margin. A 3% versus 7% revenue-growth path changes the workbench’s value-per-share output from **$173.12 to $209.00**, while a 45.5% versus 47.5% gross-margin path changes it from **$185.93 to $194.90**. This ranking is range-dependent and uses FCFE; it is decision evidence about operating-driver exposure, not the final required FCFF valuation result.

Evidence ledger and current conventions

Item	Classification	Value / unit	Period / as-of	Primary source or calculation	Treatment and uncertainty
AAPL market price	Reported fact	$337.00/share	Sep. 17, 2026 close	Price locator in `AAPL_peer_comparison_2026-09-10.md`	Used for market comparison only; verify the final price-source export before submission.
Cash plus marketable securities	Reported fact	$146.517B	Jun. 27, 2026	Apple Q3 FY2026 Form 10-Q	Included as non-operating assets in the current FCFF DCF; final policy must explain whether every marketable security is excess cash.
Commercial paper plus term debt	Reported fact	$84.344B	Jun. 27, 2026	Apple Q3 FY2026 Form 10-Q	Included in the bridge. Lease policy remains unresolved.
Diluted-share convention	Reported fact / convention	14.594B shares	Jul. 17, 2026	Apple Q3 FY2026 Form 10-Q cover page	Point-in-time shares are used in the current DCF; final work must disclose the difference from weighted-average diluted shares.
Starting FCFF	Forecast assumption / calculated input	$131.0B	TTM estimate as of Sep. 2026	`AAPL_reverseDCF_9-10-26.md`	Needs a transparent reconciliation from operating cash flow, capex, taxes, and working-capital treatment before final submission.
Explicit FCFF growth	Forecast assumption	7% annually	Years 1–5	`dcf.py`	Base case only; reverse DCF tests whether price implies a stronger path.
WACC	Forecast assumption	8%	Valuation date	`dcf.py`	Must be rebuilt from documented cost-of-capital inputs for the final model.
Terminal growth	Forecast assumption	3%	Post-Year 5	`dcf.py`	Sensitive long-run assumption; must remain below WACC and be stress-tested.
FCFF DCF value	Calculated output	$224.42/share	Sep. 17, 2026 case	`dcf.py`; `AAPL_reverseDCF_9-10-26.md`	Not an observed price; dependent on the assumptions above.
P/E cross-check	Calculated output	$206.86–$239.69/share	Sep. 17, 2026	`peer_valuation.py`; peer-comparison memo	Provisional because peer policy and comparability remain incomplete.

Validation completed and still required

Completed evidence

- The DCF runs from `python3 dcf.py` and reports its base case, sensitivity grid, and reverse-DCF result.
- The peer P/E calculation includes a manual Microsoft check: $497.75 ÷ $17.95 = 27.729805x.
- The existing sensitivity workbench records a pre-run prediction, changed input, actual result, and restored-base check in `apple_lab10_locked_record.md` and `apple_lab10_sensitivity_output.md`.
- The FCFE workbench’s accounting checks pass for the displayed sensitivity cases.

Must-fix before final Project 1 submission

1. Produce a five-year **three-statement pro-forma driven by Apple-specific inputs**, with the balance sheet balancing every year and cash flow tying to balance-sheet cash.
2. Convert/rebuild the workbench’s cash-flow output from FCFE to **FCFF**, then feed that FCFF forecast into the DCF.
3. Reconcile the $131.0B starting FCFF to filed cash-flow items and a documented tax / working-capital convention.
4. Identify and label any one-time item or basis change in the historical base year; normalize or explicitly retain it with a filing-note citation and rationale.

5. Calculate and document WACC from sourced components; retain WACC–growth sensitivity, terminal-value share, and causal low/base/high cases.
6. Set a peer-selection policy **before** treating any peer set as final; calculate at least one consistent EV multiple and one equity-value multiple.
7. Decide and document the enterprise-to-equity treatment of marketable securities, lease liabilities, pension obligations, and any other debt-like or non-common claims.
8. Build the required semi-adjustable analyst product: bounded driver controls, base reset, visible sensitivity results, and checks that fail loudly for incoherent inputs.
9. Create a new final Locked Changed-Input Record with a human-authored, AI-off prediction, pre-run timestamp and repository commit, old/new units, expected and actual direction, decision effect, and diagnosis. The existing lab record is supporting evidence but does not yet document every final-project field.
10. Run and preserve the known-answer reconciliation, independent source/definition check, highest-risk failure test, and clean-environment README-only cold run.
11. Create the final repository README, visible executed output, validation-and-AI-use record, decision memo/deck, corrected transcripts, three videos, and 12-row submission manifest.

AI-use record for this Edition B

Tool / surface	Material task	Independent check or limitation	Disposition
Codex / generative AI	Organized the research-evolution structure, located course requirements, and helped identify that the existing workbench uses FCFE rather than FCFF.	I inspected `apple_proforma.py` and compared its FCFE label/formula with the Project 1 FCFF requirement.	Accepted as a documentation and model-review aid; the required rebuild remains my responsibility.
Generative-AI skeptical-review prompt	Identified peer-comparability as the weakest supported assumption.	I checked the peer memo’s cited business-model differences and preserved the limitation rather than adding unsupported peers.	Qualified: Microsoft and Alphabet remain leads, not final peers.

Monitoring and reversal rules

- **Initiation condition:** An initiate recommendation requires either a price at or below a refreshed, validated valuation range without deteriorating business evidence, or independently sourced evidence sufficient to support a materially stronger FCFF / margin path than the current base case.
- **Downside monitoring:** Review each earnings release for Services growth, iPhone demand, Greater China performance, gross-margin movement, regulatory developments, and cash-flow conversion.
- **Model kill rule:** Do not rely on the current DCF if the FCFF starting-point reconciliation, WACC construction, enterprise-to-equity bridge, or three-statement articulation cannot be reproduced and passed under documented checks.

Working-file map

- Edition A baseline: `project-1-edition-a.md` and `project-1-edition-a.pdf`
- FCFF DCF / reverse DCF: `dcf.py` and `AAPL_reverseDCF_9-10-26.md`
- Peer cross-check: `peer_valuation.py` and `AAPL_peer_comparison_2026-09-10.md`
- Prior pro-forma / sensitivity work: `apple_proforma.py`, `apple_sensitivity_plan.md`, `apple_lab10_locked_record.md`, and `apple_lab10_sensitivity_output.md`
- Source log: `sources.md`

What changed from Edition A to Edition B, and why?

No material change in action: I remain at watch/defer. The action did not change because the new FCFF DCF and the provisional peer-multiple check both sit b

elow the September 17 market price, while material limitations remain unresolved. The evidence changed substantially: Edition A contained a research plan and operating facts; Edition B adds dated valuation outputs, a reverse-DCF test, an imperfect but transparent market cross-check, source-led risk framing, and an explicit list of model defects that must be repaired before final submission.

The most consequential change is methodological. I determined through review of the existing code and the Project 1 brief that the Apple workbench is FCFE-based, but the required valuation architecture is FCFF plus WACC and an enterprise-to-equity bridge. I therefore classify the FCFF DCF as the current valuation engine and qualify the FCFE workbench as operating-driver sensitivity evidence only. This conclusion was shaped by course instruction and model inspection; AI assisted in organizing and challenging the work, while the code and course requirements supplied the check. The final conclusion remains conditional until the FCFF pro-forma, bridge, peer policy, validation record, and reproducible product are complete.